Posterior distributions of key model parameters from the sardine assessments to be used as input for the Operating Models for the Scoping MSE

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The initial posterior distributions for the sardine assessment assuming either no stock recruitment relationship or a Beverton Holt stock recruitment relationship are given.

Keywords: assessment, posterior distributions, sardine, South Africa

Introduction

The two-stock, two-area assessment of the South African sardine population has recently been updated to take account of the revised stock structure hypothesis (de Moor 2025a,b; de Moor *et al*. 2023) and data up to the end of 2024 (de Moor 2025c). This document provides the initial posterior distributions associated with this assessment.

Methods

Two alternative assessments have been considered thus far. S_0 assumed no stock recruitment relationship, and historical October recruitments were estimated independently (de Moor 2025a). S_{BH} assumed a Beverton Holt stock recruitment relationship with $\sigma_{r,j}^2 = 5$ (de Moor 2025d). Markov chain monte carlo was run using AD Model Builder. 10 000 000 samples were generated, with a burn-in of 2 000 000. The chain was thinned by 1000 and then 1000 samples were resampled with replacement from the remaining 8 000. Full convergence diagnostics were not considered given the time constraints for the Scoping Management Strategy Evaluation, although the burn-in was selected from visual inspection of the traces.

Results and Discussion

The medians and 90%iles of the posterior distributions are given in Table 1 and Figures 1 to 6. Many parameters are estimated less precisely for S_0 than for S_{BH}. These initial runs have provided information which can help finalise the assessments for providing input to the Operating Models for the joint sardine-anchovy Management Strategy Evaluation. Alternative fixed natural mortalities and proportion of WTS which move from the west to the south coast at ages 1+ will be tested, together with a more informative prior distribution on the age at which the length is zero. The multiplicative bias associated with the hydro-acoustic survey was estimated to be larger than expected for S_0 and this will need further investigation if future estimates are similar once the above alternatives have been run and convergence has been checked.

References

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Table 1. The posterior median and 90% probability intervals for key parameters and outputs for S_0 and S_{BH}. All parameters are defined in the Appendix of de Moor (2025a), with Beverton Holt stock recruitment parameters given in de Moor (2025d).

Parameter	S_{BH}	S_0
M_{juv}^{diff}	0.28 [0.16,0.45]	0.37 [0.13,0.49]
M_{ad}^{diff}	0.50 [0.50,0.50]	0.46 [0.41,0.49]
M_{inc}^{WTS}	0.00 [0.00,0.00]	0.00 [0.00,0.00]
$move_{y,0}^{CTS}$	0.00 [0.00,0.00]	0.00 [0.00,0.00]
ϕ	1.00 [1.00,1.00]	0.99 [0.99,1.00]
$\chi_{w,y,1} = \chi_{w,1}$	0.50 [0.05,0.95]	0.50 [0.05,0.95]
$\chi_{w,y,4} = \chi_{w,4}$	0.50 [0.05,0.95]	0.50 [0.05,0.95]
$(\sigma_1^{sel})^2$	3.92 [2.23,6.20]	3.13 [2.09,7.60]
$(\sigma_2^{sel})^2$	0.78 [0.62,0.84]	0.90 [0.68,1.14]
$\bar{l}_{2,s,y\leq1998,1} - \bar{l}_{1,y}$	10.45 [9.50,11.43]	10.93 [8.65,14.35]
$\bar{l}_{2,s,y\geq1999,1} - \bar{l}_{1,y}$	10.85 [10.19,11.51]	10.96 [8.85,14.41]
$\bar{l}_{2,w,y\leq1994,1} - \bar{l}_{1,y}$	8.45 [7.03,9.39]	8.16 [4.25,11.00]
$\bar{l}_{2,w,y\geq1995,1} - \bar{l}_{1,y}$	8.46 [7.08,9.33]	8.52 [3.53,10.63]
$\bar{l}_{3,s,y\leq1998,1} - \bar{l}_{1,y}$	6.22 [5.82,6.41]	6.29 [5.21,7.47]
$\bar{l}_{3,s,y\geq1999,1} - \bar{l}_{1,y}$	7.54 [7.11,7.84]	7.28 [6.00,8.70]
$\bar{l}_{3,w,y\leq1994,1} - \bar{l}_{1,y}$	5.82 [5.52,6.19]	5.86 [4.63,7.09]
$\bar{l}_{3,w,y\geq1995,1} - \bar{l}_{1,y}$	6.05 [4.59,6.30]	4.71 [3.12,6.16]
S_{50}	8.83 [8.43,9.31]	4.96 [0.81,7.40]
δ	1.12 [0.96,1.35]	1.77 [0.19,4.09]
κ_w	1.09 [1.04,1.13]	1.71 [1.46,1.94]
$\kappa_s - \kappa_w$	0.00 [0.00,0.00]	0.00 [0.00,0.00]
κ_s	1.09 [1.04,1.13]	1.71 [1.46,1.94]
$L_{w,1}$	12.26 [12.08,12.39]	11.83 [11.29,12.32]
$L_{s,1} - L_{w,1}$	2.56 [1.59,3.05]	0.24 [0.02,0.93]
$L_{w,\infty}$	20.26 [20.03,20.84]	18.08 [17.17,18.75]
$L_{s,\infty}$	19.42 [18.71,19.67]	19.24 [18.71,19.79]
$t_{0,s}$	-0.32 [-0.38,-0.25]	0.35 [0.22,0.43]
$t_{0,w}$	0.14 [0.12,0.19]	0.44 [0.36,0.49]
ϑ_0	2.81 [2.58,2.98]	2.10 [0.48,2.91]
ϑ_1	2.49 [2.31,2.77]	2.71 [2.32,2.96]
ϑ_{2+}	0.94 [0.88,1.09]	1.19 [0.97,1.47]
ϑ_{bf}	0.11 [0.06,0.13]	0.11 [0.01,0.36]
ϑ_{af}	0.03 [0.00,0.10]	0.15 [0.01,0.48]
$ln(k_{ac})$	-0.49 [-0.70,-0.26]	-1.07 [-1.26,-0.88]
k_{ac}	0.61 [0.50,0.78]	0.34 [0.29,0.42]
k_{cov}	0.43 [0.39,0.47]	0.46 [0.38,0.55]

Table 1. (continued)

Parameter	S_{BH}	S_0
k_{covS}	0.95 [0.84,0.98]	0.26 [0.05,0.62]
K^{WTS}	2392 [768,3120]	-
K^{CTS}	195 [123,273]	-
K^{WTS}	0.38 [0.25,0.48]	-
K^{CTS}	0.50 [0.32,0.63]	-

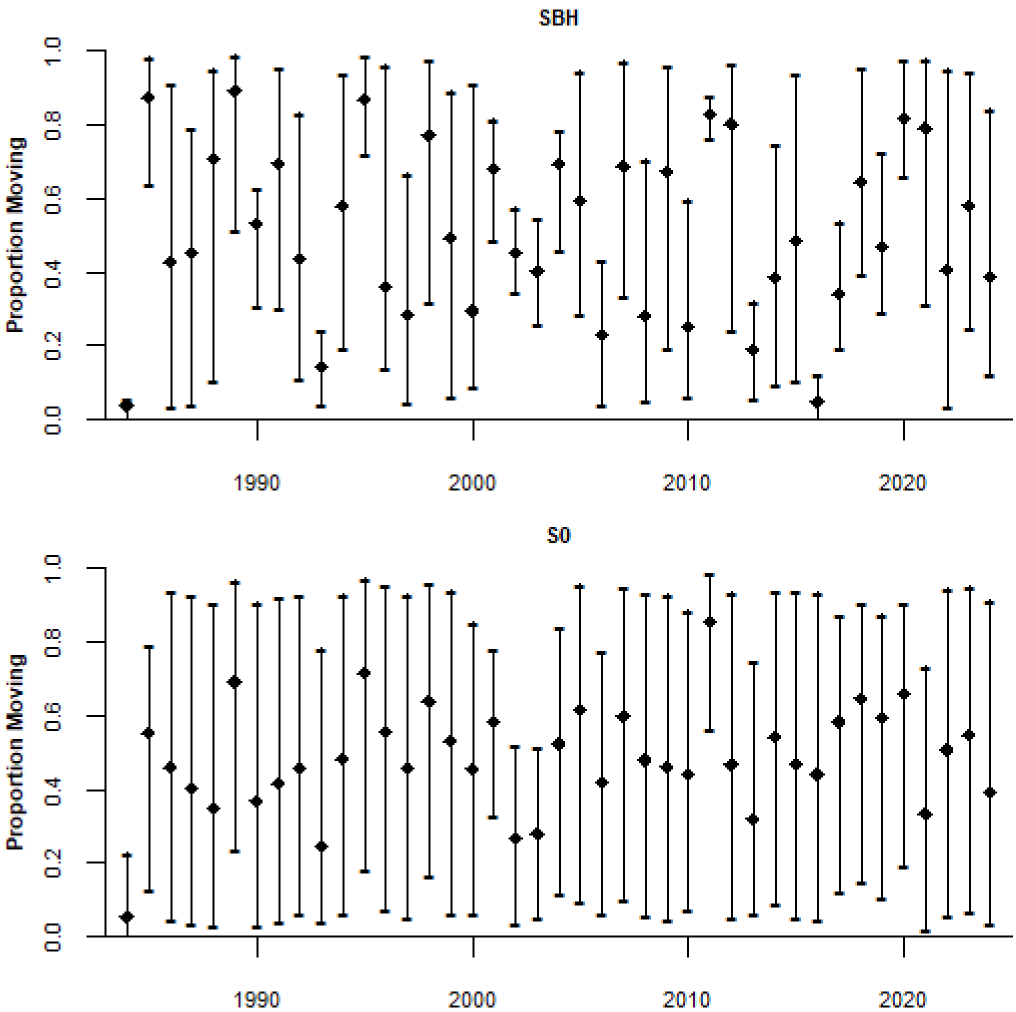

Figure 1. The median and 90% probability interval of the proportion of 0-year old WTS which move from the west to the south coast each year, $move_{y,0}^{WTS}$, for S_{BH} and S_0.

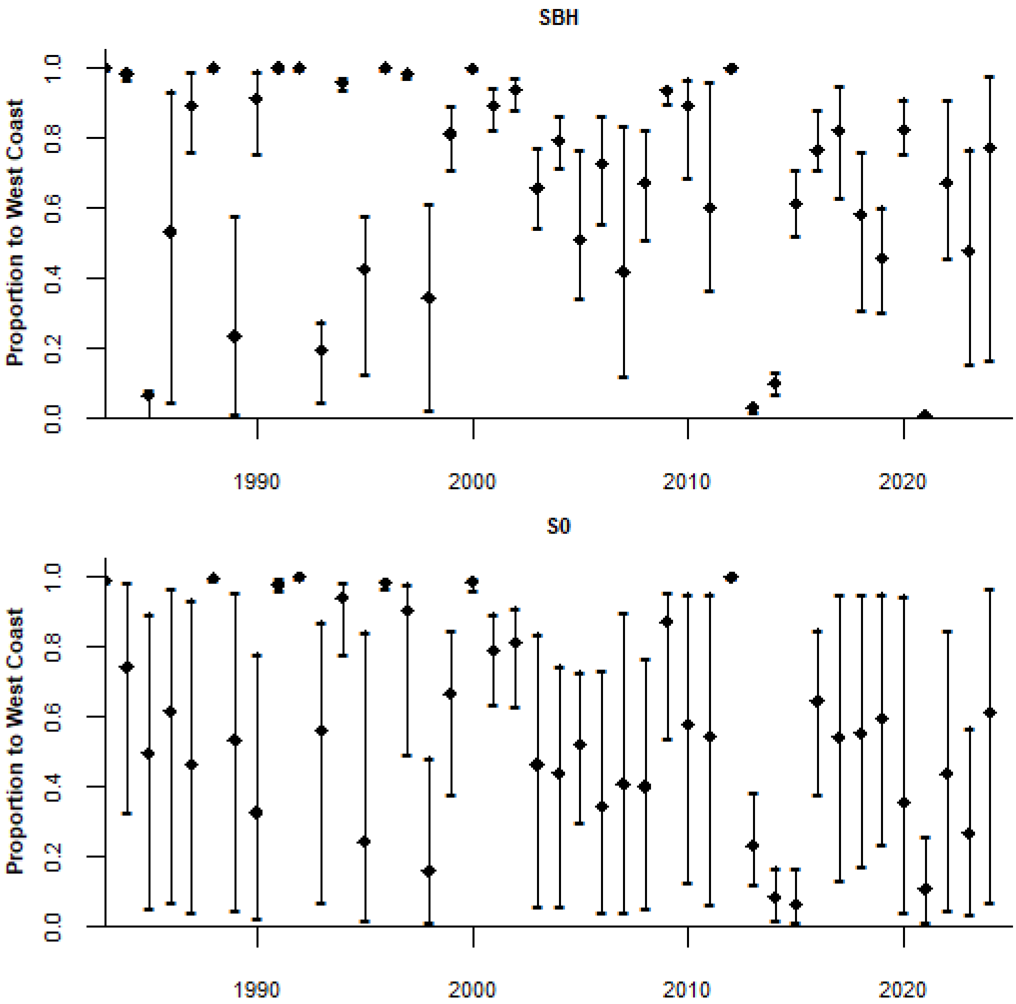

Figure 2. The median and 90% probability interval of the proportion of WTS recruits which recruit to the west coast, p_y, for S_{BH} and S_0.

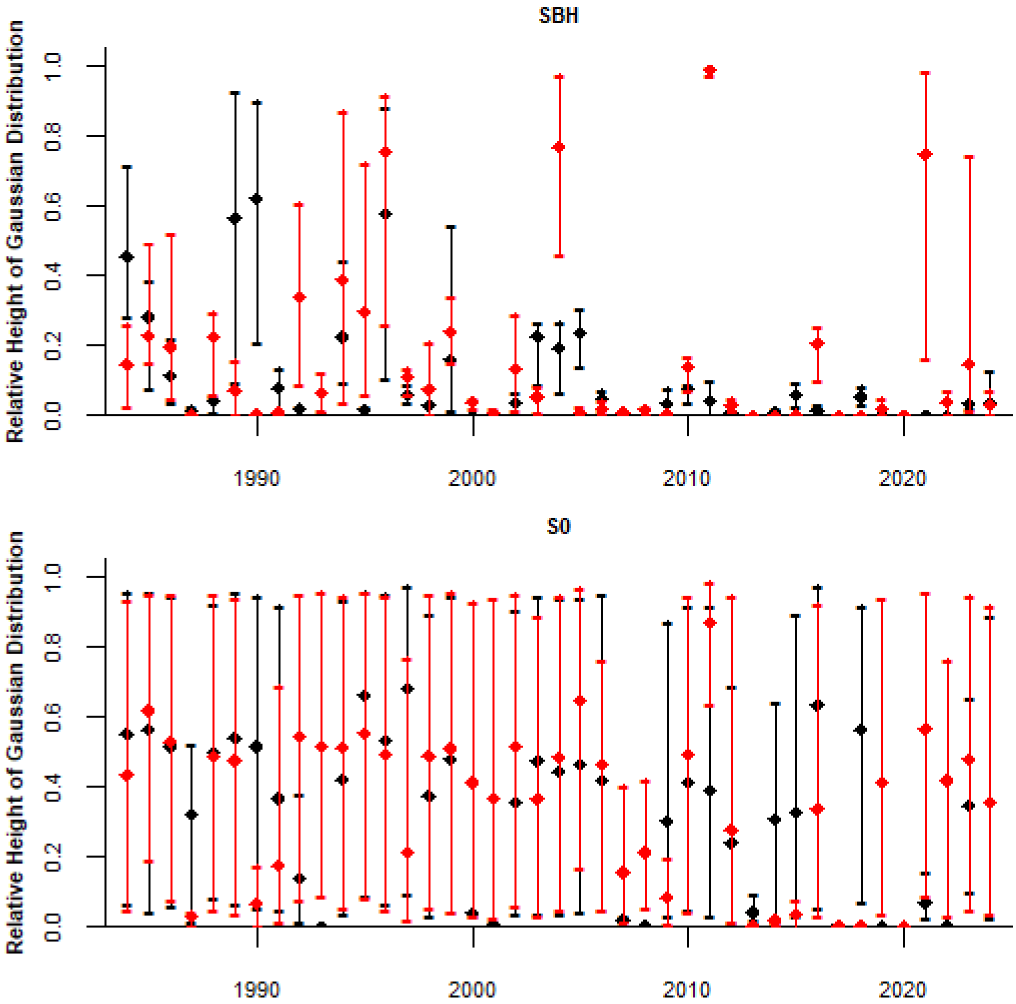

Figure 3. The median and 90% probability interval of the height of the commercial selectivity Gaussian distribution relative to the height of the logistic distribution $\chi_{w,y,q=2}$(black) and $\chi_{w,y,q=3}$ (red) for S_{BH} and S_0.

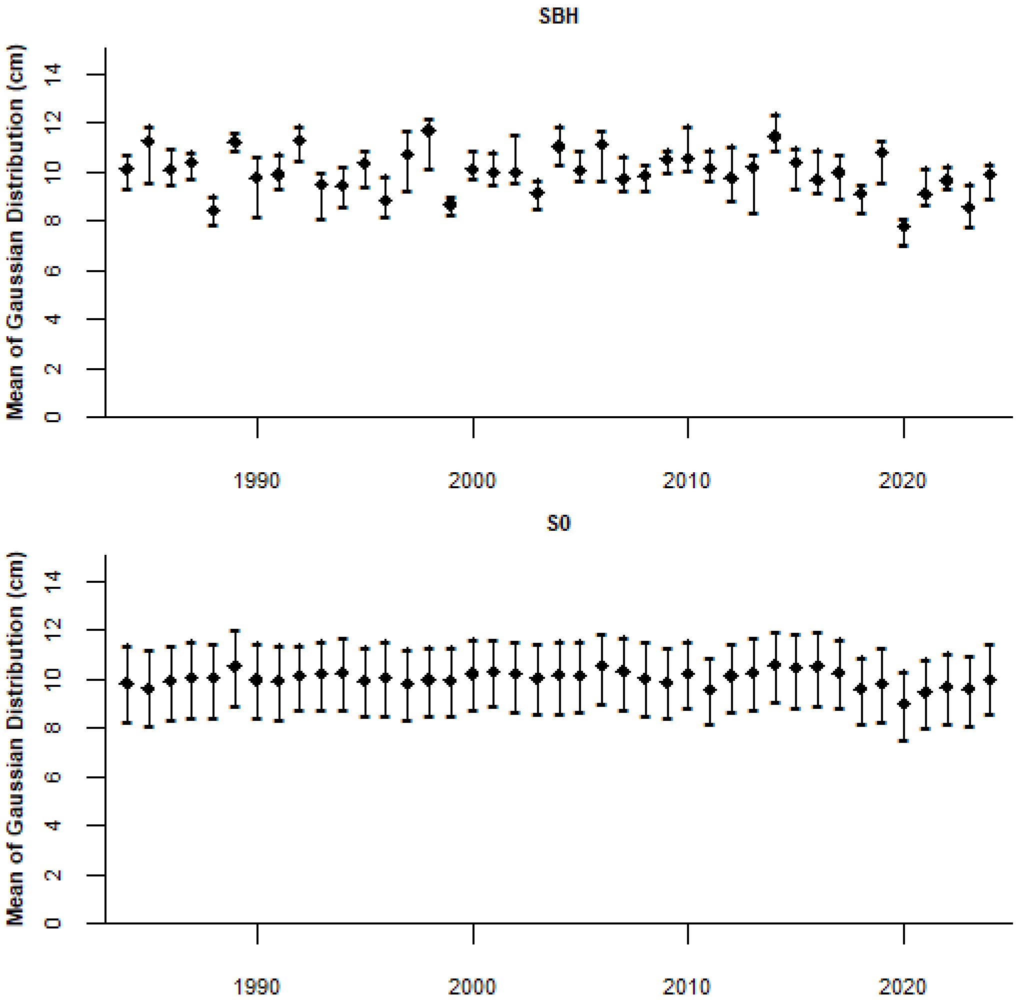

Figure 4. The median and 90% probability interval of the mean of the commercial selectivity Gaussian distribution, $\bar{l}_{1,y}$, for S_{BH} and S_0.

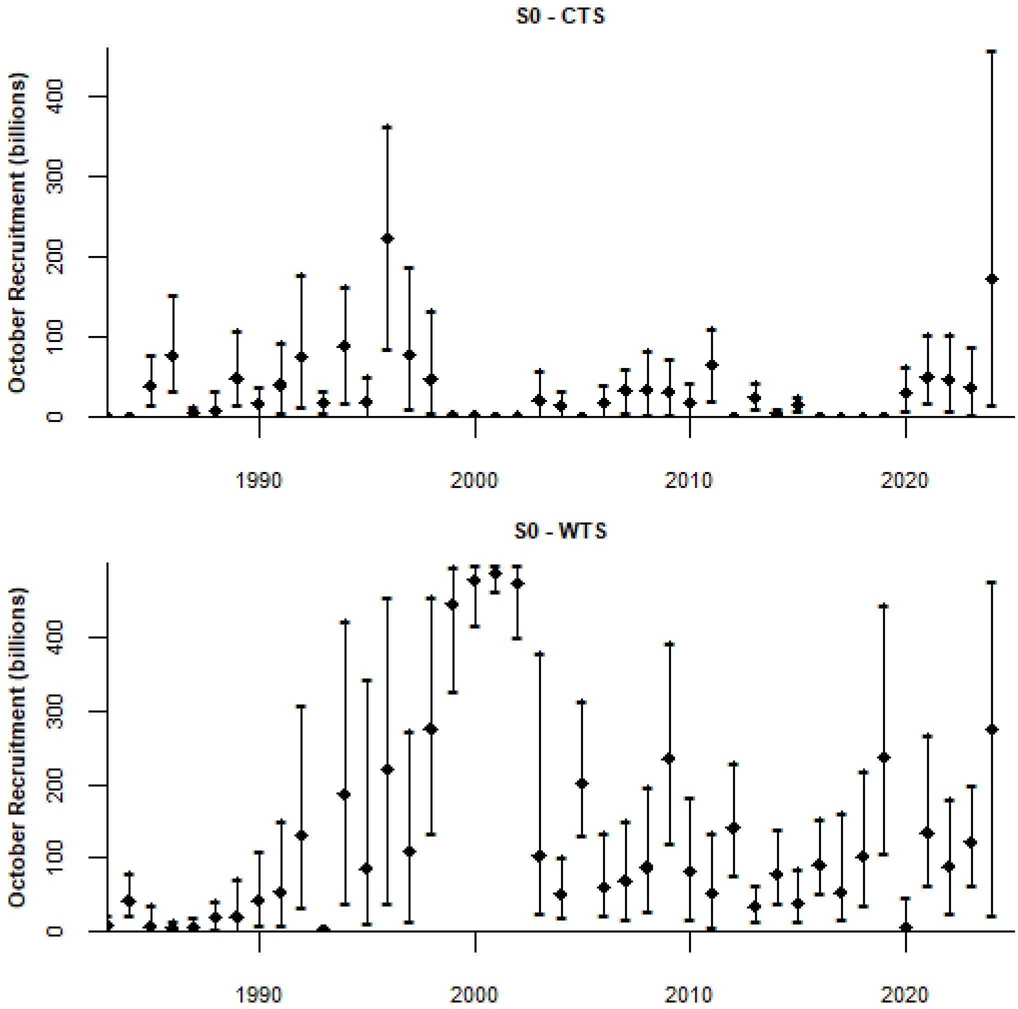

Figure 5. The median and 90% probability intervals of the annual recruitment estimated for CTS and WTS for S_0.

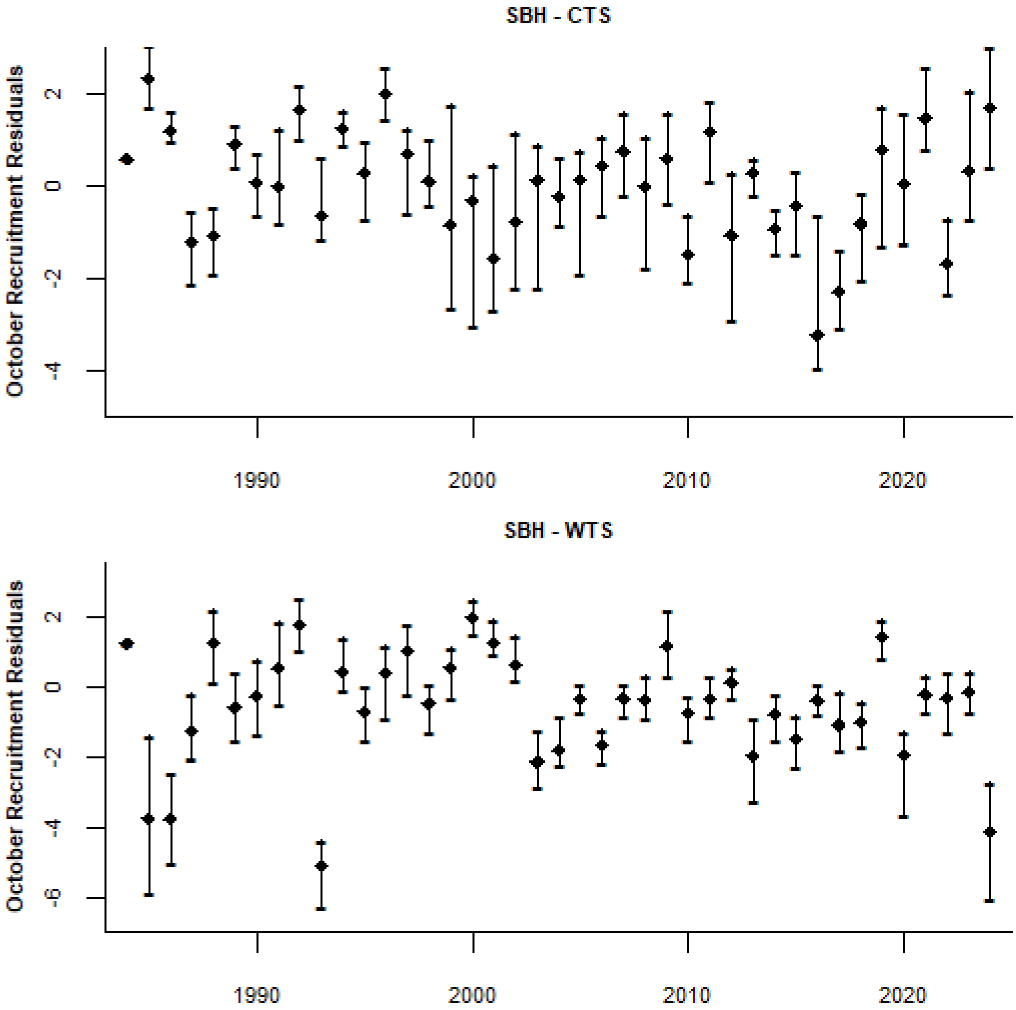

Figure 6. The median and 90% probability intervals of the annual recruitment residuals about the Beverton Holt stock recruitment relationship for S_{BH}.